

2° INTERNATIONAL
BIOHACKATHON

OF THE BIOINFORMATICS ITALIAN SOCIETY 2026

2° International BioHackathon of BITS

Padova, May 26-27 2026
In person event



Welcome & General information

Schedule

- Session 1: 26 May 2026, 14:30–18:30
- Session 2: 27 May 2026, 08:30–12:30

WiFi connection

- Option 1: **Eduroam**
- Option 2: **Ad-hoc network**
 - network: **zona01-unipd**
 - username: **BioHack2026**
 - password: **BioHack_26_27_05**

Outline

- Tasks
- Challenge
- Data
- Evaluation
- Computational Resources

Task

Develop bioinformatics pipelines for

- Task 1): Reverse engineering of bacterial network
- Task 2): Reverse engineering of gene regulatory network

Task 1) Reverse engineering of bacterial network

Participant should **develop bioinformatics pipeline** that, **starting from raw count data**, performs

- count preprocessing (e.g. normalization, zero imputation, data transformation) if needed
- reverse engineering of bacterial network

The pipeline should provide as output an ***adjacency matrix*** describing the inferred bacteria interaction graph

Task 2) Reverse engineering of gene regulatory network

Participant should **develop bioinformatics pipeline** that, **starting from raw count data**, performs

- count preprocessing (e.g. normalization, zero imputation, data transformation) if needed
- reverse engineering of gene network

The pipeline should provide as output an ***adjacency matrix*** modelling the gene network

The challenge

- Each group should perform **both tasks**
- For each task, **some test dataset with known interaction networks will be provided** (i.e. pairs of raw count data + true network)
- Groups should test the goodness of the proposed pipelines on such datasets
- A balance between the different performance metrics (e.g. precision, recall) should be considered

Data

Data are available here:

<https://github.com/younginfolife/BiohackathonBITS2026/tree/main>

For each task you have **10 test datasets**. Each dataset contains

- **raw count matrix** (features on row, samples on column, csv format).
This should be the **input of your pipeline**!
- **true adjacency matrix**. This is useful to **test the results of your pipeline**, the ideal pipeline should be able to infer it!
- **metadata/information** (for some dataset)

Data

For each task you have also **3 evaluation dataset**

Each dataset contains just the raw count matrix, and optionally some metadata

These are the datasets on which we will evaluate your pipeline, i.e. the dataset on which you should run your pipeline and send us the inferred adjacency matrices for evaluation

Evaluation

For each task, each group will be **evaluated only on the evaluation datasets**

For each task, each group should **submit**

- a **markdown of the developed pipeline**
- the **computed adjacency matrices** for each evaluation dataset, following the required format

Markdown and adjacency matrices should be submitted **here**

Computational resource

Pre-installed software for metagenomics data analyses (R packages)

- Normalization: GMPR, CSS, RCSS, TMM
- Zero imputation: scImpute, drImpute, MBimpute
- Data transformation: CLR (Microbiome)
- Reverse engineering: SPRING, SPIEC-EASI, CCLASSO

Pre-installed software for transcriptomics data analyses (R packages)

- Reverse engineering: Huge, Glasso, Genie3, Space, GeneNet, C3Net, WGCMA, Minet, qgraph, BDgraph

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Let's start!



Social event: BioHackathon Aperitif